DINESH NATARAJAN

Aerospace Engineer | Engineering Task Author for AI Training | Design, Simulation and Verification

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PROFILE

MSc aerospace engineer who builds complete engineering task packages for AI training: the problem specification, a verified reference design, and objective checks that grade any valid answer. Authored CFD and structural design tasks for frontier AI agents at Alignerr (Labelbox), each calibrated against repeated agent runs, and now reviews aerospace engineering solutions for technical accuracy at Innodata. The engineering underneath is hands-on: a twin-boom UAV taken from its requirements brief to a 3D-printed model, and a topology-optimised engine bracket (22% lighter at a factor of safety of 1.73) with full fatigue and damage-tolerance substantiation. Works in Python, MATLAB/Simulink, ANSYS, OpenFOAM, XFLR5 (XFoil), Creo Parametric and CATIA V5 on Linux.

CORE SKILLS

- **Engineering task and benchmark design:** problem specifications with defined inputs, constraints, operating conditions, output metrics and numeric pass/fail acceptance criteria; verified reference designs; deterministic Python graders and rubrics; hidden evaluation cases; difficulty calibration; loophole and exploit review
- **Aerospace design and analysis:** wing planform and airfoil selection; UAV conceptual sizing and trade studies (constraint diagram, sizing matrix, carpet plot); CG, trim and static margin; structural FEA (static, modal, contact); topology optimisation; fatigue and damage tolerance; hand-calculation cross-checks
- **CAD:** parametric part modelling in PTC Creo Parametric and CATIA V5; GD&T (ASME Y14.5); 3D-printed model build
- **Simulation and analysis:** ANSYS Mechanical with Topology Optimization, Abaqus/Standard, Altair HyperMesh, OpenFOAM, XFLR5 (XFoil), AeroSandbox (vortex lattice), OpenSeesPy, MATLAB/Simulink
- **Scripting and environment:** Python (NumPy, pandas, scikit-learn, Matplotlib), Bash, Linux (Ubuntu), Docker, Git and GitHub, CI/CD automated testing, technical reports

PROFESSIONAL EXPERIENCE

Reinforcement Learning Task Author, Engineering Simulation (Contract)
Jun 2026 to Sep 2026

Alignerr (Labelbox) | Remote

- **Task ideation and specs:** Wrote engineering design problems with defined inputs, constraints, operating conditions, output metrics and numeric acceptance criteria, framed around the required function rather than one prescribed geometry so different design approaches can pass.
- **Aerospace design task:** Wing planform task in which the agent sets aspect ratio, taper, twist and sweep to trim inside a static-margin band under a root-bending cap while maximising L/D, analysed by vortex lattice (AeroSandbox) behind an OpenFOAM solver-health gate; removed a leading-edge vs quarter-chord sweep ambiguity from the spec.
- **Reference model build:** Built a verified reference solution for each task that scores 1.000 against its own grader, and checked that trivial submissions score near zero while other strategies spread across the range (0.00 to 0.67 on one task).
- **Objective verification:** Coded deterministic Python graders that reject poor meshes, diverged solves and physically invalid results, and score the worst case across operating conditions, including hidden evaluation cases.
- **Complexity review:** Calibrated difficulty against repeated frontier-agent runs to land in a target score band (wing task retuned from 0.90 to 0.62). Closed a geometry loophole, pulled a ready-to-call model that let agents optimise straight to the reference, and caught an additive rubric floor that made the difficulty gate unreachable, then proposed multiplicative feasibility gating.
- **Agent run review:** Diagnosed failures from agent transcripts (a read-only output file that blocked 5 of 5 runs; time-outs from long Monte Carlo sweeps) and reworked instructions so runs finish inside the time limit (a 71-minute time-out became a 49-minute completion).
- **Task portfolio:** CFD and aerodynamics: wing planform design; backward-facing-step reattachment control in OpenFOAM, rebuilt after tracing an agent plateau to a contradictory spec key. Structural dynamics in OpenSeesPy: viscous damper retrofit of a soft storey; seismic retrofit of a 12-storey shear wall; tuned-mass absorber allocation under two hazard families.
- **Physics and delivery:** Replaced Rayleigh damping, which over-damped under uniform ground excitation, with explicit dashpots matching the reference damping matrix; delivered every task through Docker, GitHub pull requests and CI/CD.

Aerospace Engineering Expert, AI Training (Contract)
Sep 2026 to Present

Innodata Inc., via Upwork | Remote

- Review and rate aerospace engineering question, solution and answer sets used to train AI models, including problems built on engineering drawings.
- Check each worked solution for technical accuracy and completeness, and write a justification for every rating.

Procurement and Inventory Operations Supervisor — May 2023 to Sep 2025

Tops | Manchester, United Kingdom

- Led a 21-member operations team, running procurement, stock replenishment and inventory control.

Product Design Engineer (Intern) — Oct 2020 to Apr 2021

Black Space Research and Development | Bangarpet, Karnataka, India

- Built a shell-element FE model of a truck rear underrun protection device (RUPD) in Altair HyperMesh and solved three regulatory load cases with contact in Abaqus/Standard.
- Identified the governing case (234.6 MPa against 230 MPa yield at the mounting flange) and recommended stronger material grades.

DESIGN PROJECTS

Aircraft Engine Bracket: Topology Optimisation and Fatigue Substantiation — 2023, extended 2026

MSc dissertation, University of Salford; independent extension | PTC Creo Parametric, ANSYS Mechanical

- Modelled two bracket concepts in Creo Parametric for the GE bracket challenge loads in SAE 4130 steel; ran three static load cases plus modal analysis in ANSYS Mechanical.
- Cut mass 22% through topology optimisation while holding a factor of safety of 1.73; a mesh-independence check moved peak stress by under 1%.
- Extended the work into a 22-page substantiation report: S-N, Goodman and Miner safe-life, NASGRO-equation crack growth and MIL-HDBK-1530 inspection intervals.

Twin-Boom Surveillance UAV: Conceptual Design to 3D-Printed Model — 2022

MSc group project (Aerospace Systems Design), University of Salford

- Three-person team sized a 2.2 m, 2.8 kg twin-boom pusher UAV within a 7 kg design brief using a constraint diagram, a 3 × 3 sizing matrix and a carpet plot, with CG, trim and lateral-directional stability checks.
- Led the airfoil analysis (Eppler E193 wing, S9023 tail), the design integration and the final report.
- Led the 3D printing and control-surface integration of the model at the Salford Makerspace.

Natural-Laminar-Flow Airfoil Study — 2024

Independent project | XFLR5 (XFoil)

- Compared NACA 63-215 against NACA 2412 at cruise: peak L/D of 67.3 against 46.8 (+44%) and about 42% lower cruise drag.

Cessna 172 Longitudinal Flight Dynamics

Independent project | MATLAB/Simulink

- Built a 4-state small-perturbation state-space model (u, w, q, θ) with eigenvalue stability analysis; met the MIL-F-8785C short-period criteria (ω_n 5.60 rad/s, ζ 0.58) and passed 6 of 6 V&V checks.

EDUCATION

MSc Aerospace Engineering — 2023

University of Salford, United Kingdom

Dissertation: structural analysis of an aircraft engine bracket using the finite element method (ANSYS). Coursework: Aerospace Systems Design, Computational Fluid Dynamics, Finite Element Analysis, Structural Dynamics, Composite Structures, Additive Manufacturing.

BTech Aerospace Engineering — 2021

Karunya Institute of Technology and Sciences, Coimbatore, India

CERTIFICATIONS

- **Master Diploma in Product Design** | CADD Centre, 2022
- **Airbus Supply Chain Specialist** | Airbus Beyond, 2026
- **Machine Learning Specialization** | DeepLearning.AI and Stanford Online (in progress)